SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE Name:	Cloud Computing and BigData Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25

Question Count: 5

Code of Conduct

I **Abdul Riyaz Ahammad (192525106)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Abdul Riyaz Ahammad**

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

ANSWERS

Question 1: Auto-Scaling and Load Balancing for Seasonal Traffic

Introduction

Seasonal sales (for example, a Diwali or Black Friday sale) produce sudden and unpredictable spikes in requests. A fixed on-premises setup must be sized for the peak and sits idle the rest of the year, or it is under-sized and crashes. Cloud computing solves this through elasticity, which is delivered mainly by auto-scaling and load balancing.

How auto-scaling handles demand

- **Horizontal scaling:** An auto-scaling group (AWS Auto Scaling, Azure VM Scale Sets, Google Managed Instance Groups) adds or removes virtual machine or container instances automatically based on policies such as CPU above 70%, request count per instance, or a schedule.
- **Scheduled and predictive scaling:** Capacity is raised before a known sale starts and reduced afterwards, while target-tracking policies react to unexpected surges.
- **Scale-in:** When traffic falls, extra instances are terminated, so the company pays only for what it used (pay-as-you-go).

How load balancing helps

A load balancer (Elastic Load Balancer, Azure Load Balancer, Cloud Load Balancing) is the single entry point for users. It distributes incoming requests across all healthy instances using algorithms such as round robin, least connections or weighted routing. It performs health checks and stops sending traffic to failed instances, and it can spread instances across multiple Availability Zones. Newly launched instances are registered with the load balancer automatically, so scaling and balancing work together.

Reducing downtime and improving user experience

- No single point of failure: if one server or zone fails, traffic is rerouted to healthy servers, so the site stays available.
- Consistent response time: because capacity grows with demand, pages load quickly and checkouts do not time out during peak hours.
- Fewer abandoned carts and lost sales, which protects revenue and brand reputation.
- Rolling updates can be done behind the load balancer without taking the site offline.

Role of monitoring tools

Monitoring tools (Amazon CloudWatch, Azure Monitor, Google Cloud Monitoring, Prometheus/Grafana) collect metrics such as CPU, memory, latency, error rate and request count. These metrics trigger the auto-scaling policies, drive dashboards and alarms (SNS, email, SMS) for the operations team, and provide logs for finding bottlenecks. After the sale, the same data supports capacity planning and cost optimization.

Real-world e-commerce scenario

An online retailer such as Flipkart or Amazon runs its web and application tiers in an auto-scaling group behind an Application Load Balancer across three Availability Zones. At the start of a Big Billion Days style sale, CloudWatch detects CPU above 70% and the group grows from 10 to 80 instances within minutes. The load balancer spreads shoppers across them, product catalog pages are cached, and the database uses read replicas. When the sale ends, the group shrinks back to 10 instances. Result: no downtime, fast checkout, and costs that follow actual demand.

Conclusion

Auto-scaling supplies the capacity, load balancing distributes the work, and monitoring supplies the feedback that makes the system self-adjusting. Together they give high availability, good user experience and cost efficiency during unpredictable spikes.

Question 2: Secure Cloud Backup Storage

Introduction

Organizations need to keep large volumes of backup data for long periods without losing it or exposing it. Cloud object and archive storage (Amazon S3 and Glacier, Azure Blob Storage, Google Cloud Storage) is designed for this purpose.

Durability and data redundancy

- **Replication:** Each object is stored as multiple copies across different devices and facilities. Amazon S3, for example, stores data redundantly across at least three Availability Zones and is designed for 99.999999999% (eleven nines) durability.
- **Erasure coding and checksums:** Data is split into fragments with parity information, and checksums detect and automatically repair corruption or lost disks.
- **Redundancy options:** Locally redundant (LRS), zone-redundant (ZRS), geo-redundant (GRS) and cross-region replication let the company choose the level of protection against disk, data center or regional failure.
- **Versioning and object lock:** Earlier versions of files are retained and can be made immutable (WORM), protecting backups from accidental deletion and ransomware.

Encryption and access control

- **Encryption in transit:** TLS/HTTPS protects backups while they are uploaded or restored.
- **Encryption at rest:** AES-256 server-side encryption, with keys managed by the provider or by the customer in a key management service (AWS KMS, Azure Key Vault, Cloud KMS) or HSM. Client-side encryption lets the organization hold the keys entirely.
- **Identity and Access Management (IAM):** Least-privilege policies, role-based access control, and multi-factor authentication (MFA, including MFA delete) restrict who can read, write or delete backups.
- **Network and audit controls:** Private endpoints/VPC, bucket policies that block public access, and audit logs (CloudTrail, Azure Activity Log) record every access.

Cost advantages over traditional storage

- No upfront capital expenditure on tape libraries, disk arrays, data center space, power and cooling; the cloud uses operating expenditure (pay for what is stored).

- Unlimited, elastic capacity, so there is no over-provisioning for future growth.
- Storage tiers and lifecycle policies automatically move old backups from hot to cool to archive storage (such as Glacier Deep Archive) at a fraction of the price.
- Lower administration cost, because hardware refresh, maintenance and off-site tape transport are handled by the provider.

Disaster recovery use case

A hospital group backs up its patient records nightly to encrypted, versioned storage with cross-region replication. If a ransomware attack or a fire destroys the primary data center, the IT team restores the latest clean version in the secondary region and brings up virtual machines there from replicated images (for example with AWS Elastic Disaster Recovery or Azure Site Recovery). The recovery point objective (RPO) is minutes to hours and the recovery time objective (RTO) is much shorter than rebuilding physical infrastructure.

Conclusion

Cloud storage gives backups multi-site redundancy, strong encryption and access control, and a lower cost than traditional systems, while supporting fast disaster recovery.

Question 3: Global Deployment with CDNs

Introduction

When users are spread across continents, the physical distance between a user and the origin server adds network latency and slows the app. A Content Delivery Network (CDN) such as Amazon CloudFront, Cloudflare, Akamai or Azure CDN solves this by serving content from locations close to users.

How CDNs improve performance and reduce latency

- **Caching:** Static content (images, videos, scripts, app assets, downloads) is cached at geographically distributed points of presence, so a request is answered by a nearby server instead of travelling to the origin.
- **Shorter network path:** Fewer hops and less round-trip time mean lower latency, faster load times and fewer timeouts.
- **Origin offload:** Most requests never reach the origin, which lowers server load, bandwidth cost and the chance of overload.
- **Optimization:** Compression, HTTP/2 and HTTP/3, TLS termination at the edge, image optimization and persistent connections to the origin.
- **Security:** DDoS protection and a web application firewall (WAF) at the edge absorb attacks before they reach the application.

How edge servers enhance user experience across regions

Edge servers sit in cities and ISP networks around the world. A user in Chennai is routed (by DNS or anycast) to the nearest edge location, a user in London to another, so each gets similar speed. Edge servers keep content ready for repeated requests, can run lightweight code at the edge (Lambda@Edge, Cloudflare Workers) for personalization, authentication or A/B testing, and give regional users a smooth experience with less buffering and faster page or app start.

How cloud providers support global availability

- **Regions and Availability Zones:** The app is deployed in several regions close to users, each with multiple isolated zones.
- **Global load balancing and DNS routing:** Services such as Route 53 or Traffic Manager route users by latency, geography or health, and fail over to another region if one is down.

- **Data replication:** Multi-region databases and storage replication (DynamoDB global tables, Cosmos DB, S3 cross-region replication) keep data close to users and protect against regional failure.
- **Backbone network:** Providers own high-speed private networks between regions, which is faster and more reliable than the public internet.

Real-time streaming / gaming scenario

A video streaming service like Netflix or Hotstar stores each title in multiple bitrates and caches popular segments on CDN edge servers (Netflix places Open Connect appliances inside ISPs). A viewer in India starts a show and the video segments come from a nearby edge server, so playback starts quickly with little buffering and adaptive bitrate streaming adjusts to network conditions. For an online game, the CDN delivers patches and assets quickly, while game servers are deployed in regional data centers and players are matched to the nearest region to keep ping low.

Conclusion

CDNs and edge servers reduce latency and offload the origin, and cloud regions with global routing and replication provide high availability, so a global mobile app performs consistently for every user.

Question 4: Compliance and Data Security for a Financial Institution

Introduction

Banks and financial institutions handle sensitive data and are bound by regulations such as PCI DSS, SOX, GDPR, RBI guidelines, ISO 27001 and SOC 2. Cloud use is possible when security and compliance are built in through the shared responsibility model.

How cloud providers support compliance and auditing

- **Certifications and attestations:** Providers hold independently audited certifications (ISO 27001, SOC 1/2/3, PCI DSS, HIPAA, FedRAMP) and make audit reports available through portals such as AWS Artifact or the Azure Compliance Manager.
- **Shared responsibility model:** The provider secures the underlying infrastructure (physical data centers, hardware, hypervisor), while the institution secures its own data, identities, configurations and applications.
- **Compliance tooling:** Services such as AWS Config, Azure Policy and Security Command Center continuously check resources against standards, flag non-compliant settings and generate evidence for auditors.
- **Governance:** Resource tagging, policy enforcement, separation of duties and account structures (organizations, landing zones) help enforce controls consistently.

Role of logging, monitoring and encryption

- **Logging:** Audit logs (AWS CloudTrail, Azure Activity Log, Cloud Audit Logs) record who did what, when and from where. Logs are stored immutably and give traceability for investigations and audits.
- **Monitoring and detection:** Tools such as CloudWatch, Azure Sentinel or a SIEM correlate logs and metrics, detect anomalies such as unusual logins or large data transfers, and raise real-time alerts or automated responses.
- **Encryption:** Data is encrypted in transit (TLS) and at rest (AES-256), with keys controlled through a KMS or HSM, customer-managed keys and regular key rotation. Tokenization and masking can protect card and account numbers.
- **Access control:** IAM with least privilege, MFA and privileged access management complement these controls.

Ensuring data sovereignty

- Choose cloud regions located inside the required country or jurisdiction and keep data and backups in those regions only.
- Use policies (such as AWS Service Control Policies or Azure Policy) that block resource creation or replication outside approved regions.
- Keep encryption keys in the same jurisdiction, or use customer-managed or external keys (BYOK/HYOK) and HSMs under the institution's control.
- Use sovereign or government cloud offerings where required, and review contractual terms on data location, subcontractors and government access.
- Verify through audit logs and periodic compliance reviews that data has not moved.

Example: sensitive financial transactions

An Indian bank runs its online payments platform in a cloud region in India to meet RBI data localization rules. Card data is tokenized and encrypted with customer-managed keys in a regional HSM. Every API call and database access is logged to an immutable log store, a SIEM watches for fraud patterns and unusual access and alerts the security team, and IAM with MFA limits who can reach production data. When regulators audit, the bank produces the provider's PCI DSS attestation, its own policy compliance reports and the audit trail of the transactions.

Conclusion

By combining provider certifications, continuous logging and monitoring, strong encryption, strict access control and region-restricted deployment, a financial institution can meet regulatory and sovereignty requirements in the cloud.

Question 5: Containerization, Orchestration and Portability

Introduction

Containers (for example Docker) package an application with its libraries and dependencies and run it as an isolated process, while virtual machines (VMs) emulate complete hardware with their own operating system. Containerization is the base of modern microservice and cloud-native development.

Containers vs virtual machines

Aspect	Virtual Machine	Container
Architecture	Each VM includes a full guest OS on top of a hypervisor	Containers share the host OS kernel and isolate processes using namespaces and cgroups
Size	Gigabytes	Megabytes
Startup time	Minutes	Seconds or milliseconds
Performance	Overhead from hardware emulation and guest OS	Near-native performance, little overhead
Resource usage	High; CPU and memory are reserved for each guest OS	Low; resources are shared, so many more containers fit on one host
Isolation	Strong, since each VM has its own kernel	Lighter; the kernel is shared, so a kernel flaw affects all containers
Typical use	Running different operating systems, strong multi-tenant isolation	Microservices, CI/CD, rapid scaling

In terms of security, VMs offer stronger isolation, while containers need hardening (minimal images, non-root users, image scanning, network policies, and tools such as seccomp and AppArmor).

How orchestration (Kubernetes) simplifies deployment and scaling

- **Automated deployment:** Developers declare the desired state in YAML (Deployments, Services). Kubernetes schedules containers (pods) onto suitable nodes.
- **Scaling:** The Horizontal Pod Autoscaler adds or removes pods based on CPU or custom metrics, and the cluster autoscaler adds nodes when needed.
- **Self-healing:** Failed containers are restarted or rescheduled, and unhealthy pods are replaced automatically.

- **Rolling updates and rollbacks:** New versions are released gradually without downtime and can be rolled back if there is a problem.
- **Service discovery and load balancing:** Services get stable names and traffic is distributed across pods.
- **Configuration and storage:** ConfigMaps, Secrets and persistent volumes separate configuration and data from the container image.

Portability across cloud environments

A container image runs the same way on a developer laptop, an on-premises server or any cloud, because it contains everything the application needs, so the problem of "it works on my machine" disappears. Kubernetes is available as a managed service on every major cloud (Amazon EKS, Azure AKS, Google GKE) and on-premises, and uses the same APIs and manifests everywhere. This avoids vendor lock-in, enables hybrid and multi-cloud deployments, simplifies migration, and allows workloads to move to wherever cost, performance or compliance is best.

Example: microservices-based application

An online food delivery application is divided into independent microservices: user/authentication, restaurant catalog, order, payment, delivery tracking and notification. Each service has its own container image, codebase and database, and communicates through REST/gRPC APIs or a message queue. On Kubernetes, the order service can scale to many pods during dinner hours while the catalog service stays small. A failure in the notification service does not bring down ordering, and each team can release its service independently. The same manifests deploy to AKS or GKE without change.

Conclusion

Containers are lighter and faster than VMs, Kubernetes automates deployment, scaling and recovery, and container portability lets teams run microservices consistently across different cloud environments.